

1.4 Quiz: Arithmetic Series — Markscheme

1. [Maximum mark: 7]

$$u_4 = 17 \text{ and } u_9 = 37.$$

$$\text{(a) } u_1 + 3d = 17 \quad \text{and} \quad u_1 + 8d = 37 \quad \text{M1}$$

$$\text{Subtracting: } 5d = 20, \text{ so } d = 4 \quad \text{A1}$$

$$u_1 = 17 - 3(4) = 5 \quad \text{A1}$$

M1 is for any valid route to the two unknowns: writing both equations, using $d = \frac{u_9 - u_4}{9 - 4}$, or solving the system on the GDC. Any of these earns the method mark.

A1 for $d = 4$ and A1 for $u_1 = 5$, awarded independently.

$$\text{(b) } u_n = 5 + (n - 1)(4) = 5 + 4(n - 1) \quad \text{A1}$$

Accept the simplified form $u_n = 4n + 1$, or any other equivalent expression. The substituted form is preferred because it shows where the expression comes from.

FT: award A1 for a correct expression formed from the candidate's own u_1 and d from (a).

$$\text{(c) Set } u_n = 200: \quad 5 + 4(n - 1) = 200 \quad \text{M1}$$

$$4(n - 1) = 195$$

$$n - 1 = 48.75, \text{ so } n = 49.75 \quad \text{A1}$$

n is not a positive integer, so 200 is **not** a term of the sequence. R1

Accept the equivalent set-up $4n + 1 = 200$, giving $n = 49.75$.

Accept the alternative argument: every term $4n + 1$ is odd (the terms are 5, 9, 13, ...), and 200 is even, so 200 cannot be a term. Award M1 for identifying the parity/structure of the terms, A1 for stating that all terms are odd, R1 for the conclusion.

R1 requires the conclusion to be linked to the reason. "No, because n must be a whole number" is sufficient; a bare "no" with no reason earns no R1.

FT: award R1 for a conclusion correctly drawn from the candidate's own value of n (e.g. concluding that 200 is a term if their n came out a positive integer).

2. [Maximum mark: 6]

Row 1 has 18 seats; each row has 3 more than the one in front, so $u_1 = 18$ and $d = 3$.

$$(a) \quad u_n = 18 + (n - 1)(3) \quad \text{M1}$$

$$u_n = 18 + 3(n - 1) \quad \text{A1}$$

Accept the simplified form $u_n = 3n + 15$, or any other equivalent expression, for full marks. The substituted form is preferred because it shows where the expression comes from.

M1 is for identifying $u_1 = 18$ and $d = 3$ and using the arithmetic n^{th} -term formula, however written. A candidate who writes $u_n = 18 + 3n$ (treating row 1 as having 21 seats) earns the M1 only.

$$(b) \quad u_{12} = 18 + 3(12 - 1) = 51 \text{ seats} \quad \text{A1}$$

FT: award A1 for the value obtained by substituting $n = 12$ into the candidate's own expression from (a).

$$(c) \quad 18 + 3(n - 1) > 65 \quad \text{M1}$$

$$3(n - 1) > 47$$

$$n > 16.666 \dots \quad \text{A1}$$

The first such row is row 17. A1

Accept an equation in place of the inequality ($18 + 3(n - 1) = 65$, $n = 16.67$) provided the candidate then selects the next whole row.

Accept a table search: full marks for showing both adjacent rows, $u_{16} = 63$ (≤ 65) and $u_{17} = 66$ (> 65), and stating row 17. A single row ($u_{17} = 66$, so row 17) still earns full marks, with the feedback note that the adjacent row u_{16} is what justifies "first". See markscheme-spec §18.

$16.666 \dots$ is not an acceptable final answer — the answer must be a row number.

FT: award the final A1 for the smallest integer n correctly read off from the candidate's own boundary value.

3. [Maximum mark: 5]

The series $5 + 8 + 11 + \cdots + 62$.

(a) $d = 8 - 5 = 3$ **A1**

(b) $5 + (n - 1)(3) = 62$ **M1**

$$3(n - 1) = 57$$

$$n - 1 = 19$$

$n = 20$ terms **A1**

Accept $n = 20$ found from a GDC sequence table or by listing the terms, for full marks.

$n = 19$ counts the steps of 3 from 5 to 62, not the terms; it earns the M1 only.

FT: award M1A1 for n correctly obtained with the candidate's own d from (a).

(c) $S_{20} = \frac{20}{2}(5 + 62)$ **M1**

$S_{20} = 10(67) = 670$ **A1**

Accept $S_{20} = \frac{20}{2}(2(5) + 19(3)) = 670$, or a GDC sum.

FT: award M1A1 for the sum correctly obtained with the candidate's own n from (b).

4. [Maximum mark: 7]

$u_1 = 5$ and $d = 2$, in dollars.

$$\text{(a)} \quad u_{12} = 5 + (12 - 1)(2) \quad \text{M1}$$

$$u_{12} = \$27 \quad \text{A1}$$

Accept 27 without the dollar sign.

Using 12 in place of $(n - 1)$, giving $5 + 12(2) = 29$, earns the M1 only.

$$\text{(b)} \quad S_{12} = \frac{12}{2}(5 + 27) \quad \text{M1}$$

$$S_{12} = 6(32) = \$192 \quad \text{A1}$$

Accept $S_{12} = \frac{12}{2}(2(5) + 11(2)) = 192$, or a GDC sum.

FT: when $S_n = \frac{n}{2}(u_1 + u_n)$ is used, award M1A1 for the total correctly obtained from the candidate's own u_{12} in (a).

(c) From (b), $S_{12} = 192 < 200$.

$$S_{13} = \frac{13}{2}(2(5) + (13 - 1)(2)) \quad \text{M1}$$

$$S_{13} = \frac{13}{2}(34) = 221 > 200 \quad \text{A1}$$

The first such week is week 13. A1

Accept $S_{13} = 192 + 29 = 221$.

Accept an equation or inequality in n , e.g. $n(n + 4) = 200$ giving $n = 12.28 \dots$, provided the candidate then selects the next whole week. The A1 is then for $n = 12.28 \dots$ in place of $S_{13} = 221$.

Accept a GDC table of S_n : full marks for showing both $S_{12} = 192$ and $S_{13} = 221$ and stating week 13. A single row ($S_{13} = 221$, so week 13) still earns full marks, with the feedback note that the adjacent value $S_{12} = 192$ is what justifies "first". See markscheme-spec §18.

12.28... is not an acceptable final answer — the answer must be a week number.

FT: award the final A1 for the smallest whole week correctly read from the candidate's own totals.

5. [Maximum mark: 5]

$$u_1 = 2k, \quad u_2 = 3k + 1, \quad u_3 = 5k - 2.$$

(a) The sequence is arithmetic, so the two differences are equal.

$$(3k + 1) - 2k = (5k - 2) - (3k + 1) \quad \textbf{M1}$$

$$k + 1 = 2k - 3 \quad \textbf{A1}$$

$$k = 4 \quad \textbf{AG}$$

M1 is for any valid use of the equal-difference property, including $2u_2 = u_1 + u_3$, which gives $6k + 2 = 7k - 2$. The A1 is for a correct simplified equation in k (either of these forms).

Substituting $k = 4$ and checking that the differences are equal ($13 - 8 = 18 - 13 = 5$) does not show that k must be 4; it earns the M1 only.

$$\textbf{(b)} \quad u_1 = 8, \quad u_2 = 13, \quad u_3 = 18 \quad \textbf{A1}$$

All three terms required for the mark. The value $k = 4$ is given, so a candidate who did not complete (a) can still earn this mark.

$$\textbf{(c)} \quad d = 13 - 8 = 5$$

$$S_{10} = \frac{10}{2}(2(8) + (10 - 1)(5)) \quad \textbf{M1}$$

$$S_{10} = 5(16 + 45) = 305 \quad \textbf{A1}$$

Accept the route $u_{10} = 8 + 9(5) = 53$, then $S_{10} = \frac{10}{2}(8 + 53) = 305$, or a GDC sum.

Using nd in place of $(n - 1)d$, giving $5(16 + 50) = 330$, earns the M1 only.

FT: award M1A1 for the sum correctly obtained from the candidate's own terms in (b).

6. [Maximum mark: 5]

$$u_1 = 4 \text{ and } S_{10} = 175.$$

$$\text{(a)} \quad \frac{10}{2}(2(4) + (10 - 1)d) = 175 \quad \text{M1}$$

$$5(8 + 9d) = 175$$

$$8 + 9d = 35 \quad \text{A1}$$

$$9d = 27$$

$$d = 3 \quad \text{A1}$$

The first A1 is for any correct equation once the substitution is simplified (e.g. $40 + 45d = 175$, $45d = 135$, or $9d = 27$); it does not have to be in this exact form.

Accept the route through u_{10} : $\frac{10}{2}(4 + u_{10}) = 175$ (M1), $u_{10} = 31$ (A1), then $4 + 9d = 31$, giving $d = 3$ (A1).

Using nd in place of $(n - 1)d$, i.e. $5(8 + 10d) = 175$ giving $d = 2.7$, earns the M1 only.

A correct substituted equation solved on the GDC, giving $d = 3$, earns full marks. $d = 3$ with no working seen earns the final A1 only.

$$\text{(b)} \quad S_{20} = \frac{20}{2}(2(4) + (20 - 1)(3)) \quad \text{M1}$$

$$S_{20} = 10(8 + 57) = 650 \quad \text{A1}$$

Accept $u_{20} = 4 + 19(3) = 61$, then $S_{20} = \frac{20}{2}(4 + 61) = 650$, or a GDC sum.

FT: award M1A1 for the sum correctly obtained with the candidate's own d from (a).